

UPPER RESPIRATORY TRACT INFECTION (URTI)

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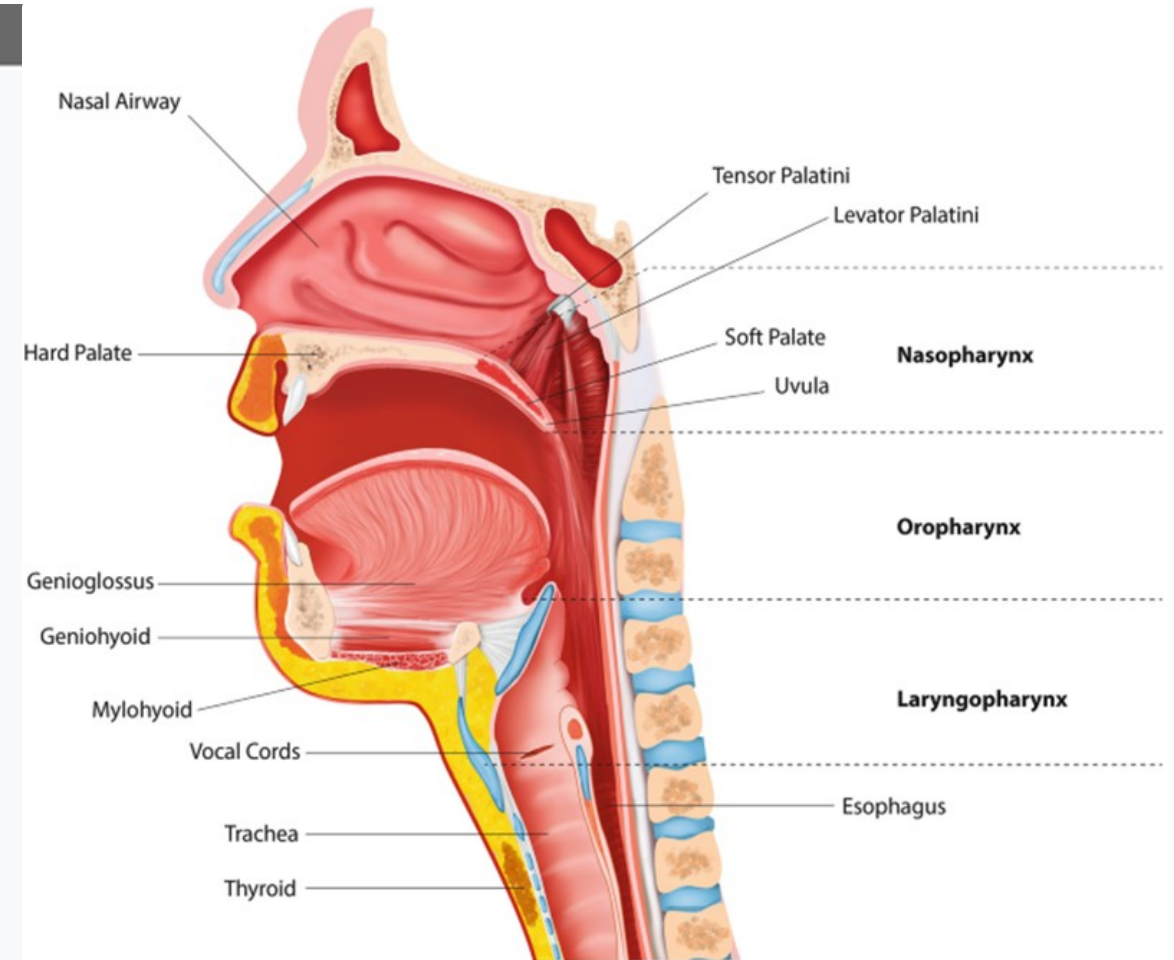
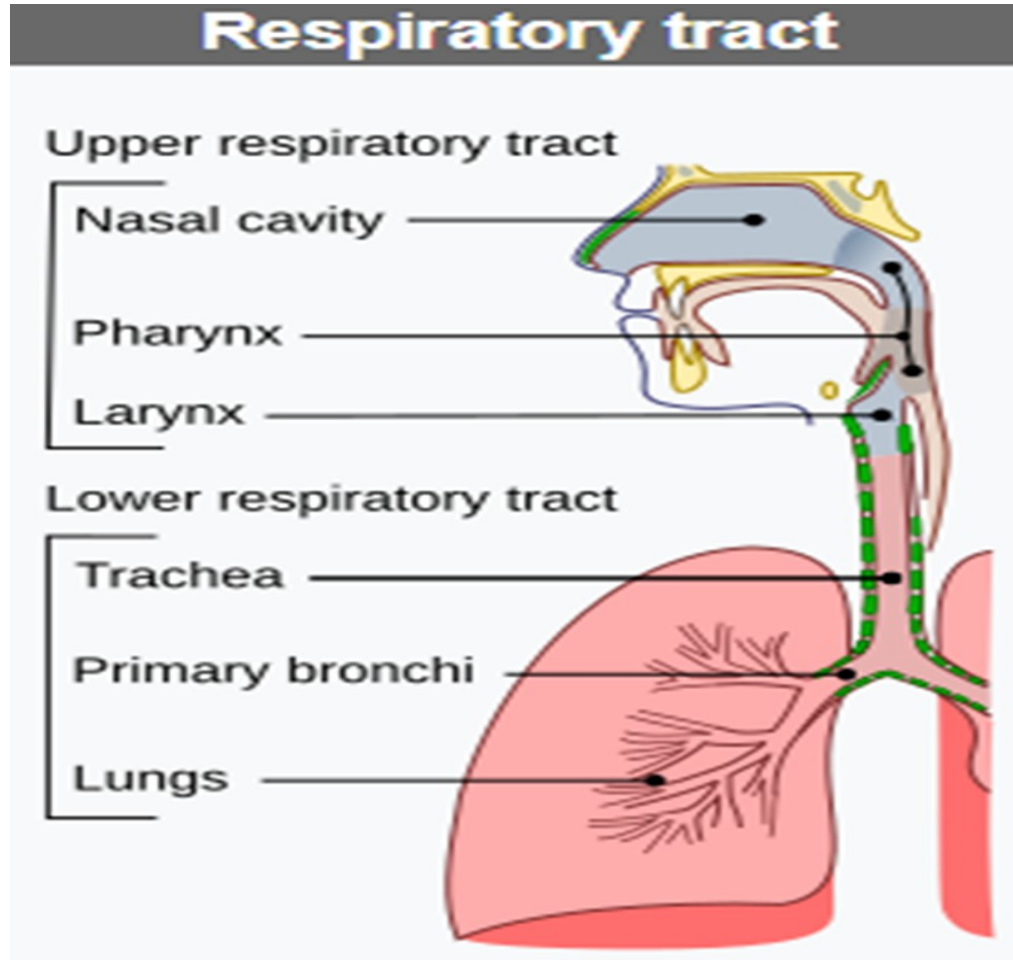
Outline

- Define upper air way
- Discussion of common upper airway infection
 - Common cold
 - Sinusitis
 - Pharyngitis(acute tonsillopharyngitis)
 - Acute ear infection
 - Epiglottitis
 - Laryngitis

Objective

- ❖ At the end of this session student able to
 - Define upper air way
 - List common URTI with their etiology
 - Describe clinical manifestation of common URTI
 - List DDX of URTI

Anatomy and definition of Upper air way



Upper respiratory tract infection

- Common URTI include
 - ✓ Rhinitis (common cold),
 - ✓ Sinusitis,
 - ✓ Acute ear infections(AOM)
 - ✓ Acute pharyngitis or tonsillopharyngitis,
 - ✓ Epiglottitis, and
 - ✓ Laryngitis(Croup)

Common cold (Rhinosinusitis)

- The common cold is an acute, self-limiting viral infection of the upper respiratory tract
- Frequently referred to as infectious rhinitis but may also include self-limited involvement of the sinus mucosa and is more correctly termed rhinosinusitis.

ETIOLOGY

Epidemiologic and clinical features of viruses that cause the common cold in children

Virus	Proportion of cases	Predominant months of circulation*	Other clinical syndromes
Rhinoviruses (more than 100 serotypes)	30 to 50%	Present year round with a peak in September and a smaller peak in March and April	
Respiratory syncytial virus	5%	November to March	▪ Bronchiolitis
Influenza viruses	5 to 15%	Winter months, with peak in February	▪ Influenza ▪ Croup ▪ Pneumonia
Parainfluenza viruses	5%	September to January with peak in October and November	▪ Croup
Adenoviruses	<5%	September to May	▪ Pharyngoconjunctival fever
Enteroviruses (echoviruses and coxsackieviruses)	<5%	Present year round, with a peak during the summer	▪ Aseptic meningitis (nonpolio enteroviruses) ▪ Herpangina (coxsackievirus type A)
Severe acute respiratory syndrome coronavirus 2	Unknown	Present year round	▪ COVID-19
Common cold coronaviruses	10 to 15%	November to February	▪ Pneumonia ▪ Croup
Human metapneumovirus	Unknown	Late winter and early spring	▪ Pneumonia ▪ Bronchitis

Epidemiology

- Colds occur year-round, but the incidence is greatest from the early fall until the late spring.
- Young children have an average of 6-8 colds per year, but 10–15% of children have at least 12 infections per year.
- The incidence of infection is primarily a function of exposure to the virus
- The incidence of illness decreases with increasing age, with two to three illnesses per year by adulthood

Pathogenesis

- **Transmission** – Viruses that cause colds are spread by three mechanisms;
 - I. **Hand contact** – Self-inoculation of one's own conjunctivae or nasal mucosa after touching a person or object contaminated with cold virus
 - II. **Inhalation of small particle droplets** (droplet nuclei or aerosols) that become airborne from sneezing or coughing
 - III. **Deposition of large particle droplets** that are expelled during sneezing and land on nasal or conjunctival mucosa (typically requires close contact with an infected person)

Cont...

- **Incubation period** – The incubation period (time between contact with infectious material until the onset of symptoms) for most common cold viruses is 24 to 72 hours.
- **Period of infectivity** – Rhinovirus shedding peaks on the third day after inoculation; this coincides with a peak in symptoms.
- Low levels of viral shedding may persist for up to two weeks.

Cont...

- The respiratory viruses have evolved different mechanisms to avoid host defenses.
 - Antigenic diversity,
 - Mucosal immunoglobulin A (IgA) induced by previous infection is short lived,
 - The brief incubation period of these viruses allows the establishment of infection before immune memory is able to respond.

CLINICAL MANIFESTATIONS

- Sore throat (typically an early manifestation),
- Fever
- Nasal congestion
- Runny nose
- Cough (2/3)
- Sneezing
- Hoarseness, headache, irritability, difficulty sleeping, decreased appetite, anterior cervical adenopathy, and conjunctival injection.

DIAGNOSIS

- The diagnosis of the common cold is made clinically, based upon history and examination findings.
- Routine laboratory studies are not helpful for the diagnosis and management of the common cold.

Differential diagnosis

Table 428.2 Conditions that Can Mimic the Common Cold	
CONDITION	DIFFERENTIATING FEATURES
Allergic rhinitis	Prominent itching and sneezing, nasal eosinophils; Hansel stain can aid diagnosis
Vasomotor rhinitis	May be triggered by irritants, weather changes, spicy foods, etc.
Rhinitis medicamentosa	History of nasal decongestant use
Foreign body	Unilateral, foul-smelling secretions Bloody nasal secretions
Sinusitis	Presence of fever, headache or facial pain, or periorbital edema or persistence of rhinorrhea or cough for longer than 10-14 days
Streptococcosis	Mucopurulent nasal discharge that excoriates the nares, no cough
Pertussis	Onset of persistent or severe paroxysmal cough
Congenital syphilis	Persistent rhinorrhea with onset in the first 3mo of life

Treatment

□ Supportive Care and Symptomatic Treatment

COMPLICATIONS

- Acute otitis media (5-30%)

- Sinusitis (5-9%)

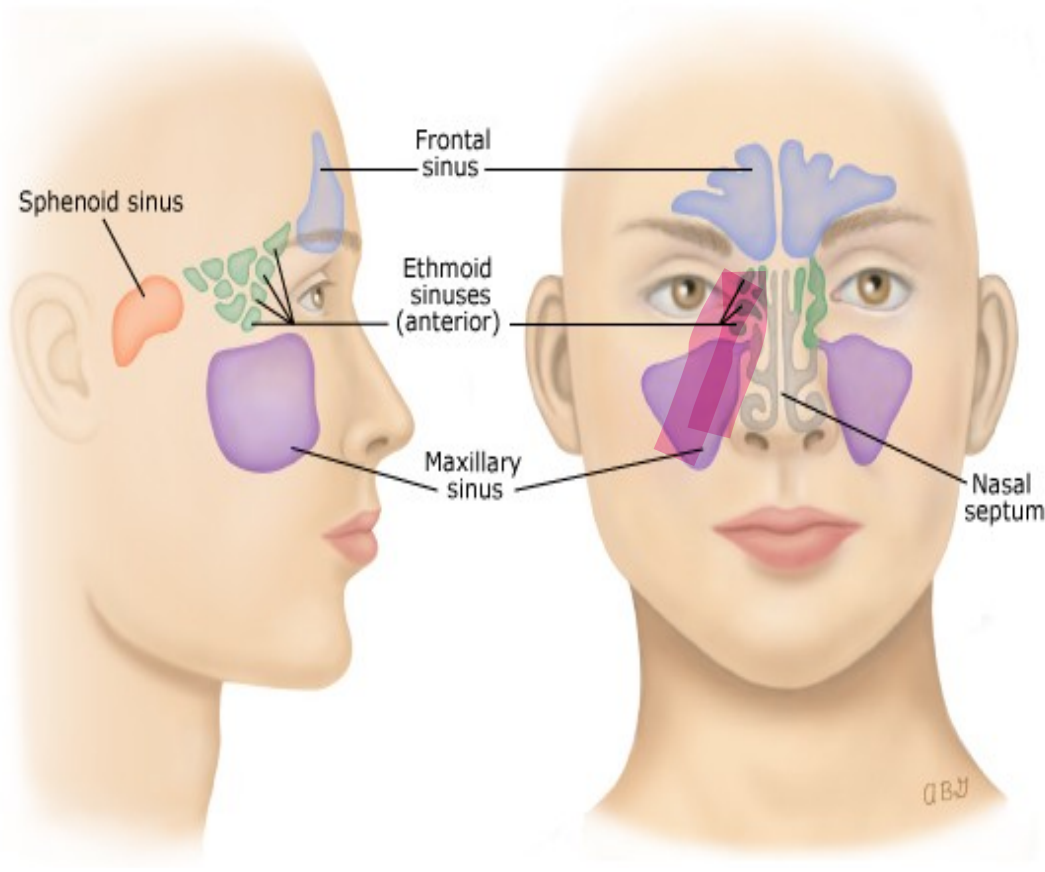
- Pneumonia

- Exacerbation of Asthma

PREVENTION

- Immunization or chemoprophylaxis against influenza
- Frequent handwashing and avoiding touching one's mouth, nose, and eyes

Sinusitis(Acute rhinosinusitis)



- ❑ Ethmoidal and maxillary sinuses are present at birth, but only the ethmoidal sinuses are pneumatized.
- ❑ The maxillary sinuses are not pneumatized until 4 years of age.
- ❑ The sphenoidal sinuses begin at 2yr pneumatized by 5 years of age,
- ❑ Frontal sinuses begin development at age 7-8 years and are not completely developed until adolescence

Pneumatized = filled with air

Definition

❑ **Sinusitis**; is inflammation of the mucosal lining of one or more of the paranasal sinuses

- The terms "sinusitis" and "rhinosinusitis" often are used interchangeably because inflammation of the paranasal sinuses is almost always accompanied by inflammation of the nasal mucosa.
- Inflammation of the sinuses is common during uncomplicated upper respiratory infections but usually resolves spontaneously.

Etiology

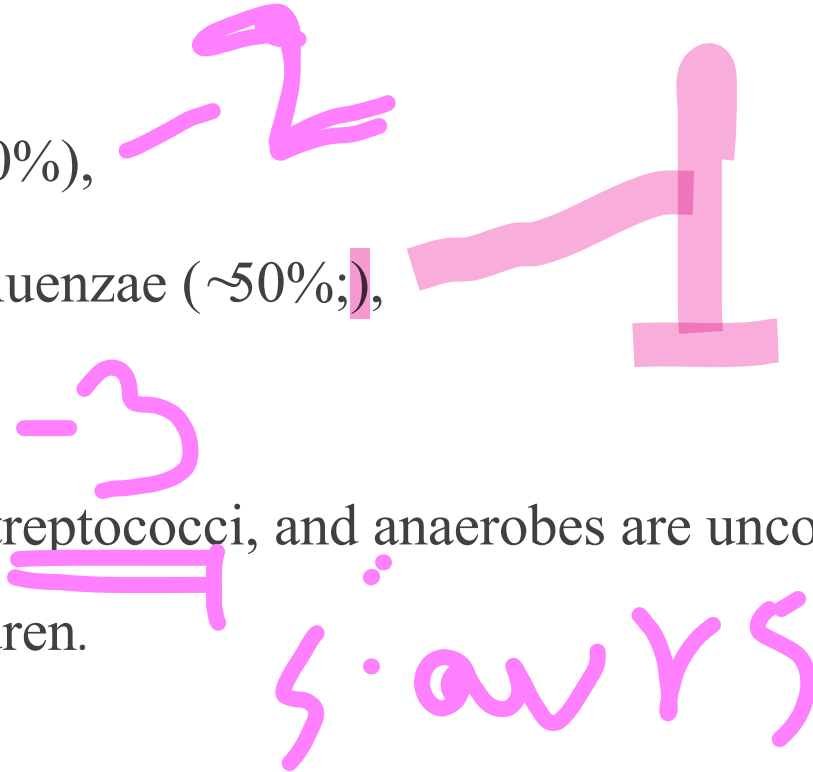
- Viral
- Bacterial (acute bacterial rhinosinusitis)
 - ✓ Acute (< 4week)
 - ✓ Subacute(4wk—3month)
 - ✓ Chronic (> 3month)
- Fungal (rare)



Acute Bacterial Rhinosinusitis

- The bacterial pathogens causing acute bacterial sinusitis in children and adolescents include

- *Streptococcus pneumoniae* (~30%),
- Non-typeable *Haemophilus influenzae* (~50%),
- *Moraxella catarrhalis* (~10%).
- *Staphylococcus aureus*, other streptococci, and anaerobes are uncommon causes of acute bacterial sinusitis in children.



EPIDEMIOLOGY

- Acute bacterial sinusitis can occur at any age and it's a common problem.
- Approximately 6 to 9 % of viral upper respiratory infections in children are complicated by the development of secondary ABRS.
- In children, ABRS typically occurs at 4-7 years of age, but it can occur at any age, even infancy.
- ABRS is less common in children younger than two years of age than in older children

Because some sinuses, especially the maxillary and frontal sinuses, are not yet fully developed.

Predisposing conditions

- Viral upper respiratory tract infections
- Allergic rhinitis,
- Tobacco smoke exposure.
- Children with immune deficiencies, particularly of antibody production (immunoglobulin [Ig]G, IgG subclasses, IgA),
- Cystic fibrosis,
- Ciliary dysfunction
- Abnormalities of phagocyte function,
- Gastroesophageal reflux,
- Anatomic defects (cleft palate), nasal polyps,
- Cocaine abuse
- Nasal foreign bodies (including nasogastric tubes), can develop chronic or recurrent sinus disease.
- Immunosuppression

PATHOGENESIS

- The paranasal sinuses are usually sterile
- Membranes that line the nose are continuous with the membranes that line the sinus cavities
- The contaminating bacteria are typically removed by mucociliary clearance.
- When mucociliary clearance is altered, the sinuses may be inoculated with large numbers of microorganisms, and infection may develop.
- Mucociliary clearance is altered by conditions that damage the ciliary epithelium or affect the number or function of cilia, the production or viscosity of mucous, or the patency of the ostia (eg, upper respiratory infections, allergic rhinitis)

CLINICAL MANIFESTATIONS

Table 429.1

Symptoms of Acute Bacterial Sinusitis

MAJOR SYMPTOMS

Purulent anterior nasal discharge
Purulent or discolored posterior nasal discharge
Nasal congestion or obstruction
Facial congestion or fullness
Facial pain or pressure alone or exacerbated by bending forward
Hyposmia or anosmia
Fever (for acute sinusitis only)

MINOR SYMPTOMS

Headache
Ear pain, pressure, or fullness
Halitosis
Dental pain
Cough
Fever (for subacute or chronic sinusitis)
Fatigue

Clinical course

- Has two clinical syndromes

1. Uncomplicated Viral upper respiratory infection
2. Acute bacterial rhinosinusitis

Clinical presentation	Description
Persistent symptoms	Nasal discharge or cough or both for >10 days without improvement
Severe symptoms	Onset with temperature of $\geq 39^{\circ}\text{C}$ (102.2°F) and purulent nasal discharge for ≥ 3 consecutive days
Worsening symptoms	Respiratory symptoms (nasal discharge or cough, or both) that worsen after initial improvement, or
	Onset of new fever or severe headache

Worsening / double sickening:

→ Gets better → then becomes worse again

DIAGNOSIS

- Uncomplicated ABRS in children is usually made clinically.
- We use both of the following criteria for diagnosis
 - I. Symptoms and signs compatible with sinus inflammation (daytime cough, nasal symptoms, or both)
 - II. Clinical course suggestive of bacterial rather than viral infection, including
 - Symptoms present **without improvement** for >10 and <30 days, or
 - Severe symptoms (ill appearance, temperature $\geq 39^{\circ}\text{C}$ (102.2°F), and purulent nasal discharge for ≥ 3 consecutive days), or
 - Worsening symptoms (increase in respiratory symptoms, new onset of severe headache or fever, or recurrence of fever after initial improvement)

NB; Imaging and microbiologic identification for complicated ABRS

DDX

- Viral upper respiratory tract infection
- Allergic rhinitis
- Nasal foreign body
- Infected adenoids
- GER

Complication of ABRS

ORBITAL

- Preseptal (periorbital) cellulitis
- Orbital cellulitis
- Orbital subperiosteal abscess
- Orbital abscess

CRANIAL

- Septic cavernous sinus thrombosis
- Meningitis
- Osteomyelitis of the frontal bone associated with a subperiosteal abscess (Pott puffy tumor)
- Epidural abscess
- Subdural abscess
- Brain abscess

TREATMENT

- ❖ Supportive care
- ❖ Symptomatic treatment
- ❖ Antibiotics therapy

Prevention

- Frequent handwashing and avoiding persons with colds.
- Vaccination
- Chemoprophylaxis

Acute Pharyngitis

- Pharyngitis is the inflammation of the mucous membranes of the oropharynx.
- Causes can be broadly categorized as infectious (usually viral(50-80%) or bacterial(GAS =5-36%)) and noninfectious.
- The incidence peaks in childhood and adolescence with approximately 50 percent of all cases occurring before age 18

INFECTIOUS ETIOLOGIES

Table 430.1

Infectious Agents that Cause Pharyngitis

VIRUSES

Adenovirus
Coronavirus including COVID-19
Cytomegalovirus
Epstein-Barr virus
Enteroviruses
Herpes simplex virus (1 and 2)
Human immunodeficiency virus
Human metapneumovirus
Influenza viruses (A and B)
Measles virus
Parainfluenza viruses
Respiratory syncytial virus
Rhinoviruses

BACTERIA

Streptococcus pyogenes (group A streptococcus)
Arcanobacterium haemolyticum
Fusobacterium necrophorum
Corynebacterium diphtheriae
Neisseria gonorrhoeae
Group C streptococci
Group G streptococci
Francisella tularensis
Yersinia pestis
Chlamydophila pneumoniae
Chlamydia trachomatis
Mycoplasma pneumoniae
Mixed anaerobes (Vincent angina)

GAS pharyngitis

- GAS is the most common cause of bacterial pharyngitis in children and adolescents.
- It accounts for 15 to 30 percent of all cases of pharyngitis in children between the ages of 5 and 15 years
- Streptococcal pharyngitis is relatively uncommon before 2-3 years of age,
- Illness occurs throughout the year but is most prevalent in winter and spring.
- It is readily spread among siblings and schoolmates.

Pathogenesis

- Transmission occur through saliva and nasal secretion which facilitated via close contact.
- Colonization of the pharynx by GAS can result in either asymptomatic carriage or acute infection.
- After an incubation period of 2-5 days, pharyngeal infection with GAS classically presents as rapid onset of significant sore throat and fever

Table 430.2

**Epidemiologic and Clinical Features
Suggestive of Group A Streptococcal and
Viral Pharyngitis**

FEATURE, BY SUSPECTED ETIOLOGIC AGENT

Group A Streptococcal

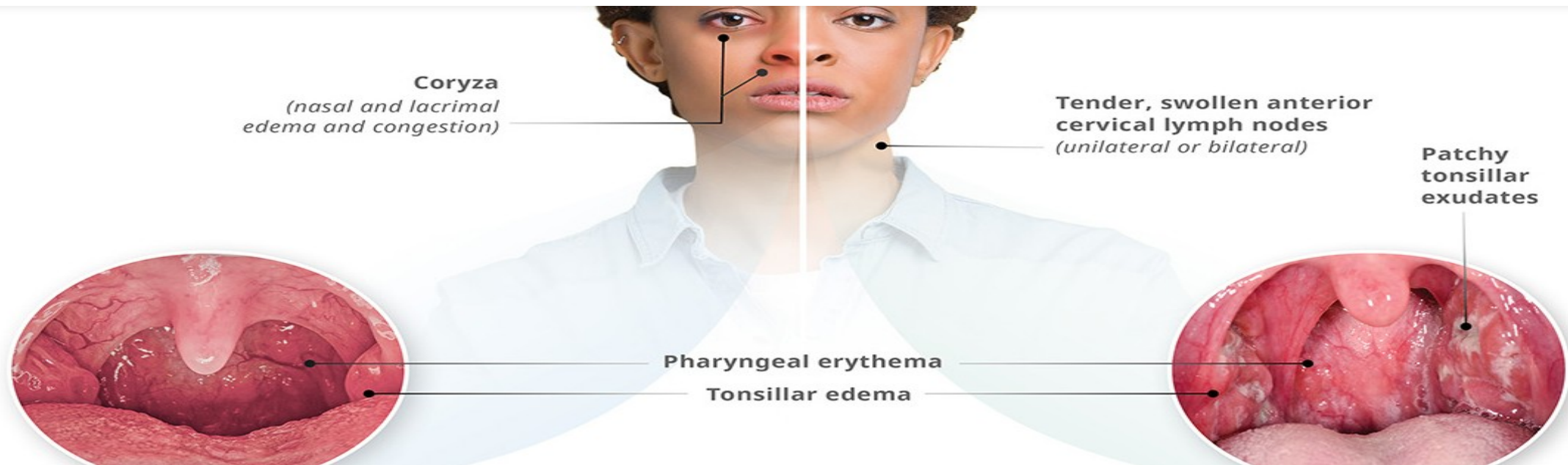
- Sudden onset of sore throat
- Age 5-15 yr
- Fever
- Headache
- Nausea, vomiting, abdominal pain
- Tonsillopharyngeal inflammation
- Patchy tonsillopharyngeal exudates
- Palatal petechiae
- Anterior cervical adenitis (tender nodes)
- Winter and early spring presentation
- History of exposure to strep pharyngitis
- Scarletiform rash
- No cough or coryza

Viral

- Conjunctivitis
- Coryza
- Cough
- Diarrhea
- Hoarseness
- Discrete ulcerative stomatitis
- Viral exanthema
- As part of the spectrum of COVID-19 and MIS-C



Fig. 431.1 Pharyngotonsillitis. This common condition has a number of causative pathogens and a wide spectrum of severity. **A**, The diffuse tonsillar and pharyngeal erythema seen here is a nonspecific finding that can be produced by a variety of pathogens. **B**, This intense erythema, seen in association with acute tonsillar enlargement and palatal petechiae, is highly suggestive of group A β -streptococcal infection, though other pathogens can produce these findings. **C**, This picture of exudative tonsillitis is most commonly seen with either group A streptococcal or Epstein-Barr virus infection. (B courtesy Michael Sherlock, MD, Lutherville, MD. From Yellon RF, McBride TP, Davis HW. *Otolaryngology*. In Zitelli BJ, Davis HW, eds. *Atlas of Pediatric Physical Diagnosis*, 4th ed. Philadelphia: Mosby; 2002:852.)



DIAGNOSIS

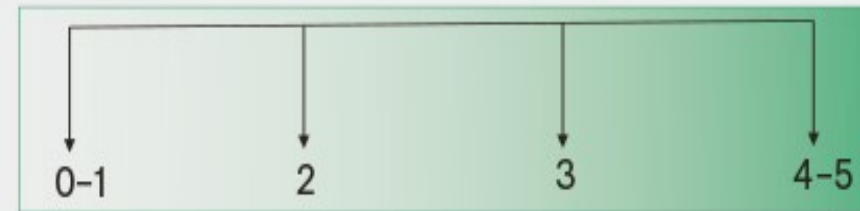
- The clinical presentations of streptococcal and viral pharyngitis often overlap.
- Based on clinical presentation and physical finding
- The diagnosis of GAS pharyngitis is supported by a positive microbiologic test (throat culture, rapid antigen detection test [RADT], or molecular point-of-care test [POC] for GAS) in a patient with symptoms of GAS pharyngitis and absence of signs and symptoms of viral infections

Table 1. McIsaac/modified Centor score

Criteria	Score
Age 3–14 years	1
Tonsillar exudate	1
Tender or swollen lateral cervical lymph nodes	1
Temperature > 38 °C	1
Absence of cough	1

Sore throat presumed infectious
(without high-risk sexual behaviours)

Calculate Modified
Centor Criteria score



No
antibiotics*

Consider:

- Rapid antigen detection test (RADT) ± throat swab to facilitate treatment for high-risk patients†
- No antibiotics*

Consider:

- No antibiotics*
- RADT ± throat swab to facilitate delayed antibiotics or failed treatment
- Delayed antibiotics (ie deterioration or no improvement in three days)*
- Treat with antibiotics if risk factors present†
- Consider Epstein-Barr virus serology as well as full blood count and liver function test (to rule out infectious mononucleosis)

Any of the
following →

- Suppurative complications
- Systemically unwell
- Scarlet fever
- Patients aged 2–25 years with sore throat in communities with a high incidence of acute rheumatic fever (eg Aboriginal and Torres Strait Islander communities in central and northern Australia, Maori and Pacific Islander people)
- Patients of any age with existing rheumatic fever/valvular heart disease

Yes ↓

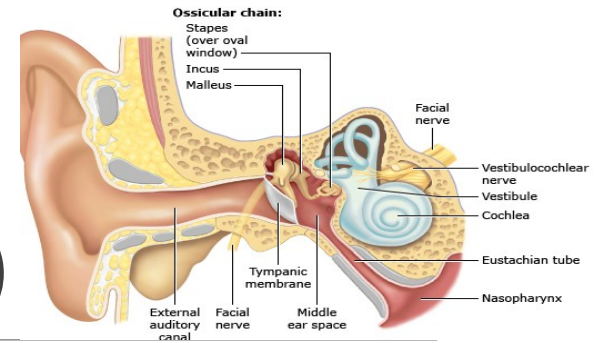
Treat with antibiotics ±
admit to hospital

Treatment

GOALS OF TREATMENT

- Reducing symptom severity and duration
- Prevention of acute complications, such as otitis media, peritonsillar abscesses, or other invasive infections
- Prevention of delayed complications or immune sequelae, particularly acute rheumatic fever, APSGN, reactive arthritis
- Prevention of spread to others

Acute ear infection(Otitis Media)



- Acute otitis media is defined as an infection of the middle ear space.
- It is a spectrum of diseases that includes
 - I. Acute otitis media (AOM),
 - II. Chronic suppurative otitis media (CSOM), and
 - III. Otitis media with effusion (OME).
- Acute otitis media is the second most common pediatric diagnosis in the emergency department, following upper respiratory infections.

Cont...

- AOM occur at any age, but commonly seen between the ages of 6 to 24 months.
- More than 80% of children experience at least one episode of OM by the age of 3 years
- Accurate diagnosis of AOM in infants and young children may be difficult
- Symptoms may not be apparent, especially in early infancy and in chronic stages of the disease

Risk and protective factors

RISK

Age
Family history
Daycare
Tobacco smoke and air pollution exposure
Pacifier use
Race & ethnicity
Socio economic status
Season of yr
Immunodeficiency

PROTECTIVE FACTORS

Breast feeding
- Vaccination

Etiology

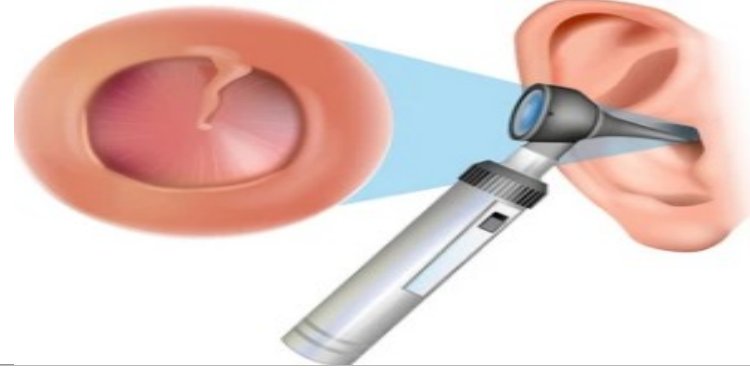
- Three pathogens predominate in AOM:
 - *Streptococcus pneumoniae*
 - Non-typeable *Haemophilus influenzae*
 - *Moraxella catarrhalis*
- Of these viruses, rhinovirus and respiratory syncytial virus are found most often

Pathogenesis

- The patient has an antecedent event (usually a viral upper respiratory tract infection) while colonized with an otopathogen(s).
- The event results in inflammatory edema of the respiratory mucosa of the nose, nasopharynx, and Eustachian tube.
- Inflammatory edema obstructs the narrowest portion of the Eustachian tube (the isthmus).
- Obstruction of the isthmus causes poor ventilation and negative middle ear pressure(leads to the accumulation of secretions produced by the middle ear mucosa).
- Viruses and bacteria that colonize the upper respiratory tract enter the middle ear via aspiration, reflux, or insufflation.

Clinical manifestation

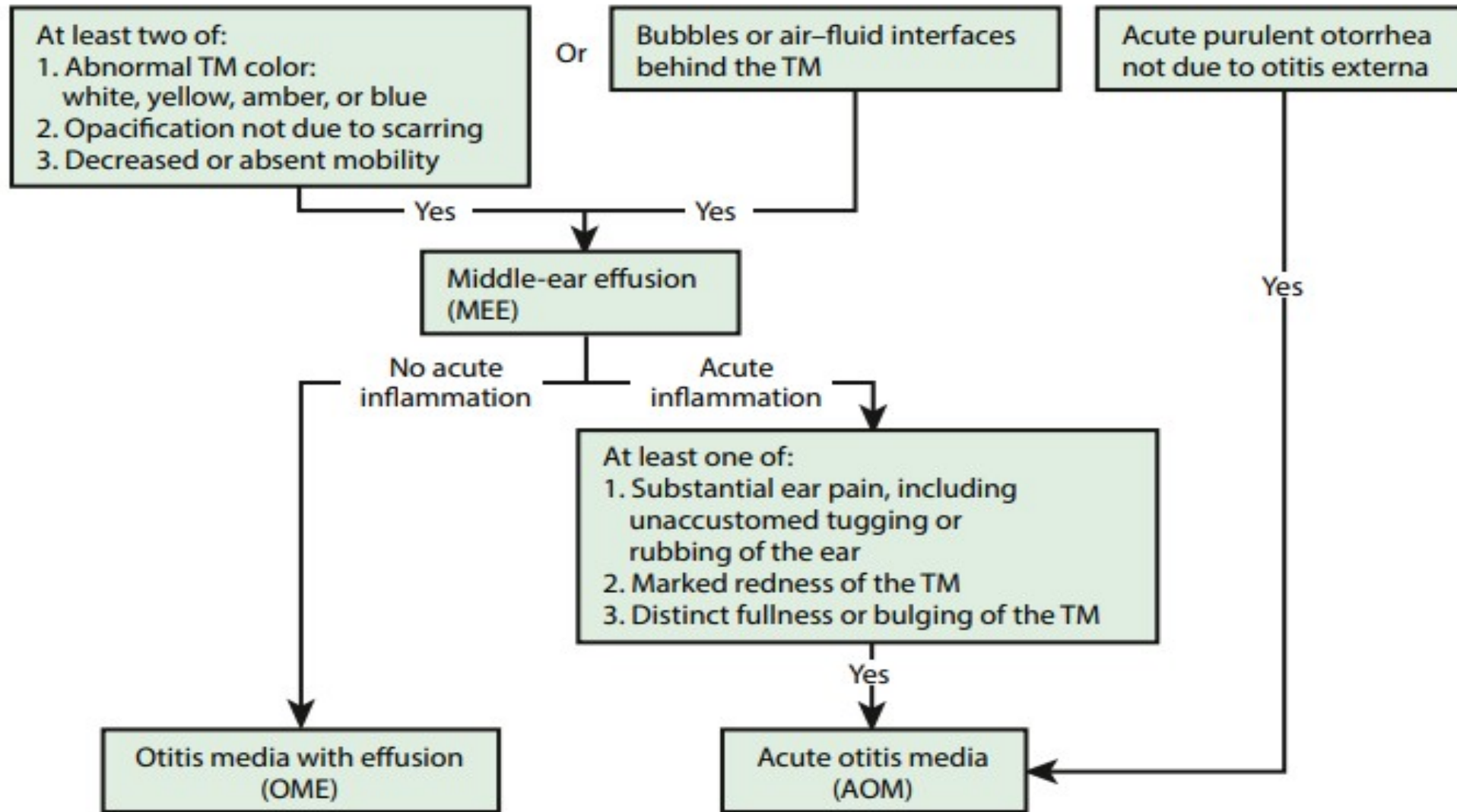
- Ear pain, ear rubbing, hearing loss, ear drainage & fever.
- Young children with AOM, particularly infants, may present with nonspecific symptoms and signs (eg, fever, fussiness, disturbed or restless sleep, poor feeding/anorexia, vomiting, diarrhea)



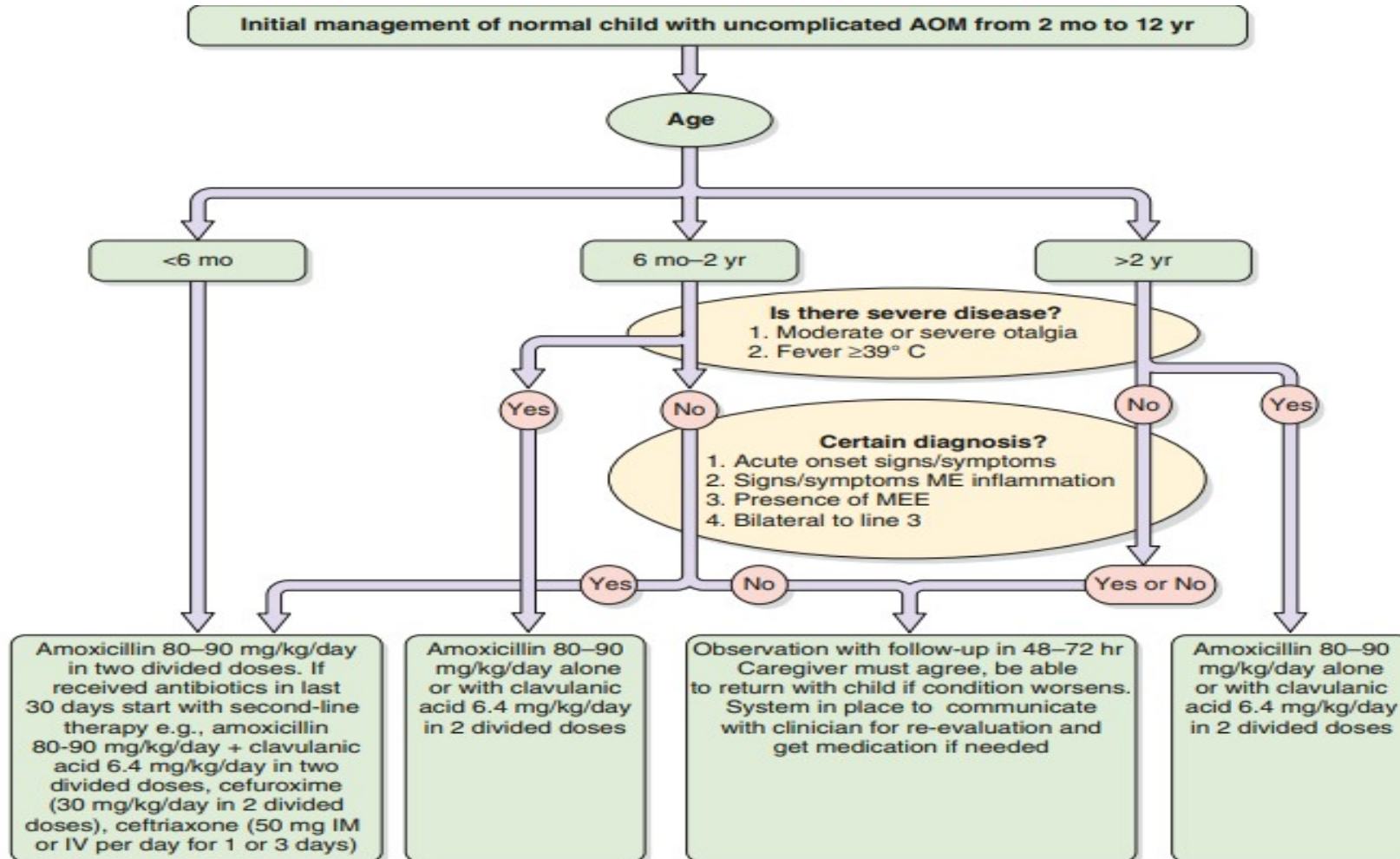
Diagnosis

- A diagnosis of AOM should be made in children who present with:
 - ✓ Moderate to severe bulging of the TM or new-onset otorrhea not caused by otitis externa
 - ✓ Mild bulging of the TM and recent (<48hr) onset of ear pain or intense TM erythema





Treatment approach



Complications and sequelae

Intratemporal complications

Hearing loss

Balance and motor problems

Tympanic membrane abnormalities

Chronic suppurative otitis media

Mastoiditis

Petrositis

Labyrinthitis

Facial paralysis

Intracranial complications

Meningitis

Epidural abscess

Brain abscess

Lateral sinus thrombosis

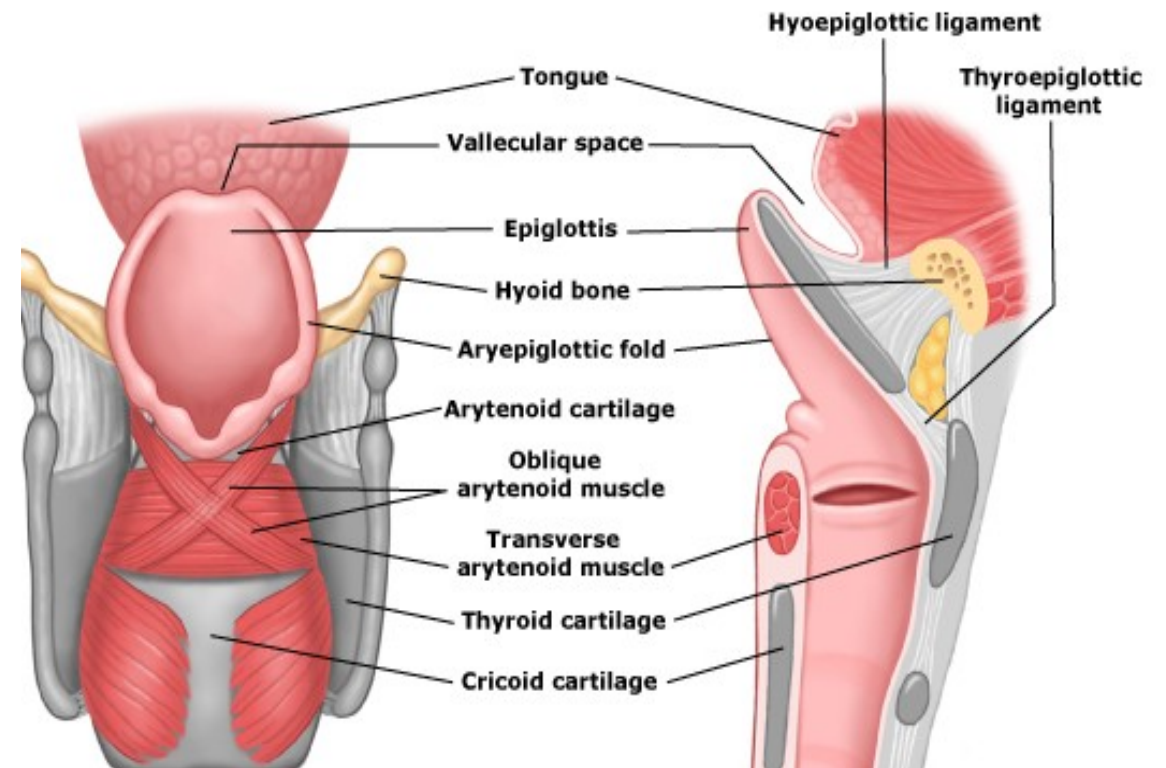
Cavernous sinus thrombosis

Subdural empyema)

Carotid artery thrombosis

Acute Epiglottitis

- Epiglottitis refers to inflammation of the epiglottis and adjacent supraglottic structures, primarily due to infection.
- Without treatment, epiglottitis can progress to life-threatening airway obstruction.



Etiology

INFECTIOUS

➤ Bacterial pathogens

- Haemophilus influenzae type b (Hib)
- Staphylococcus aureus
- Streptococcus pneumoniae ,Streptococcus pyogenes and other streptococci
- Neisseria meningitidis
- Pasteurella multocida

➤ Viral

- Influenza, type a ,Influenza, type b ,Herpes simplex virus, types 1 and 2 ,Parainfluenza virus, type 3,Epstein-Barr virus ,Human immunodeficiency virus (HIV) ,SARS-CoV-2.

➤ Fungal(candida albican)

NON INFECTIOUS

❖ Local airway trauma such as:

- Thermal injury from:
 - -Ingestion of hot beverages or food
 - -E-cigarette use or inhalation of heated objects during use of illicit drugs such as crack cocaine
- Direct trauma to the epiglottis
- Caustic ingestion or inhalation
- Lymphoproliferative disease or GVH-disease after bone marrow or solid organ transplantation
- Chronic granulomatous disorders

Clinical presentation

ACUTE (FEVER AND STRIDOR)

- Toxic appearance and distress (agitation, restlessness, irritability Sudden onset of high fever (between 38.8 and 40.0°C)
- Stridor
- Drooling
- Change in voice (muffled, "hot potato" voice)
- Severe sore throat and dysphagia
- "tripod" posture

SUBACUTE (SEVERE SORE THROAT)

- Progressively painful sore throat
- Low-grade fever
- Muffled ("hot potato") or hoarse voice
- Difficulty with swallowing
- Drooling



This child's "tripod" positioning (trunk leaning forward, neck hyperextended, chin thrust forward) is caused by epiglottitis and represents the patient's attempt to maximize the patency of a significantly obstructed upper airway. Also, note the child's toxic appearance.

This 4-year-old girl has epiglottitis caused by *Haemophilus influenzae* type b.

(A) She prefers to sit and appears anxious.

(B) The child assumes the characteristic sniffing position to maximize the patency of her airway.

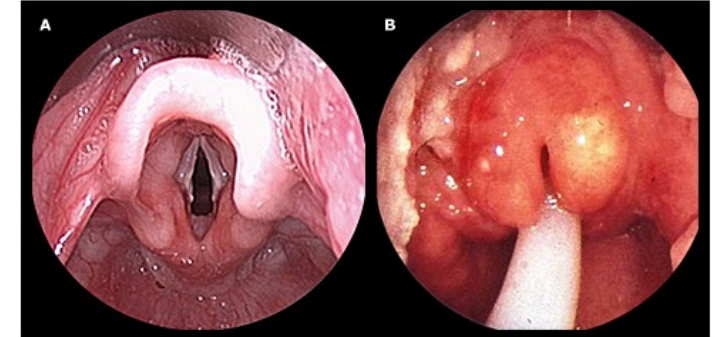
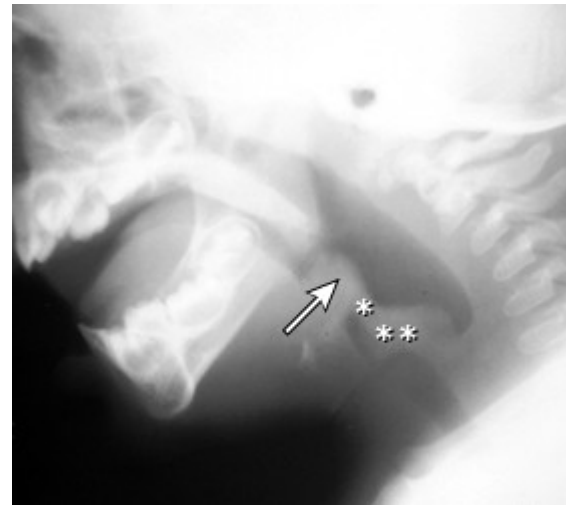
DIAGNOSIS

Clinical suspicion — Epiglottitis typically presents in one of two ways:

- **Progressive, severe sore throat over several days**
- **Abrupt onset of fever, stridor, and respiratory distress over 24 hours**
 - • "Tripod" position ,Anxiety ,Sore throat ,Stridor,Drooling,Dysphagia ,Severe respiratory distress

Diagnostic confirmation

- Visualization of the epiglottis
- Plain radiographs
- Laboratory
- Microbiology



(A) Normal epiglottis.

(B) Characteristic erythematous, edematous epiglottis of acute epiglottitis.

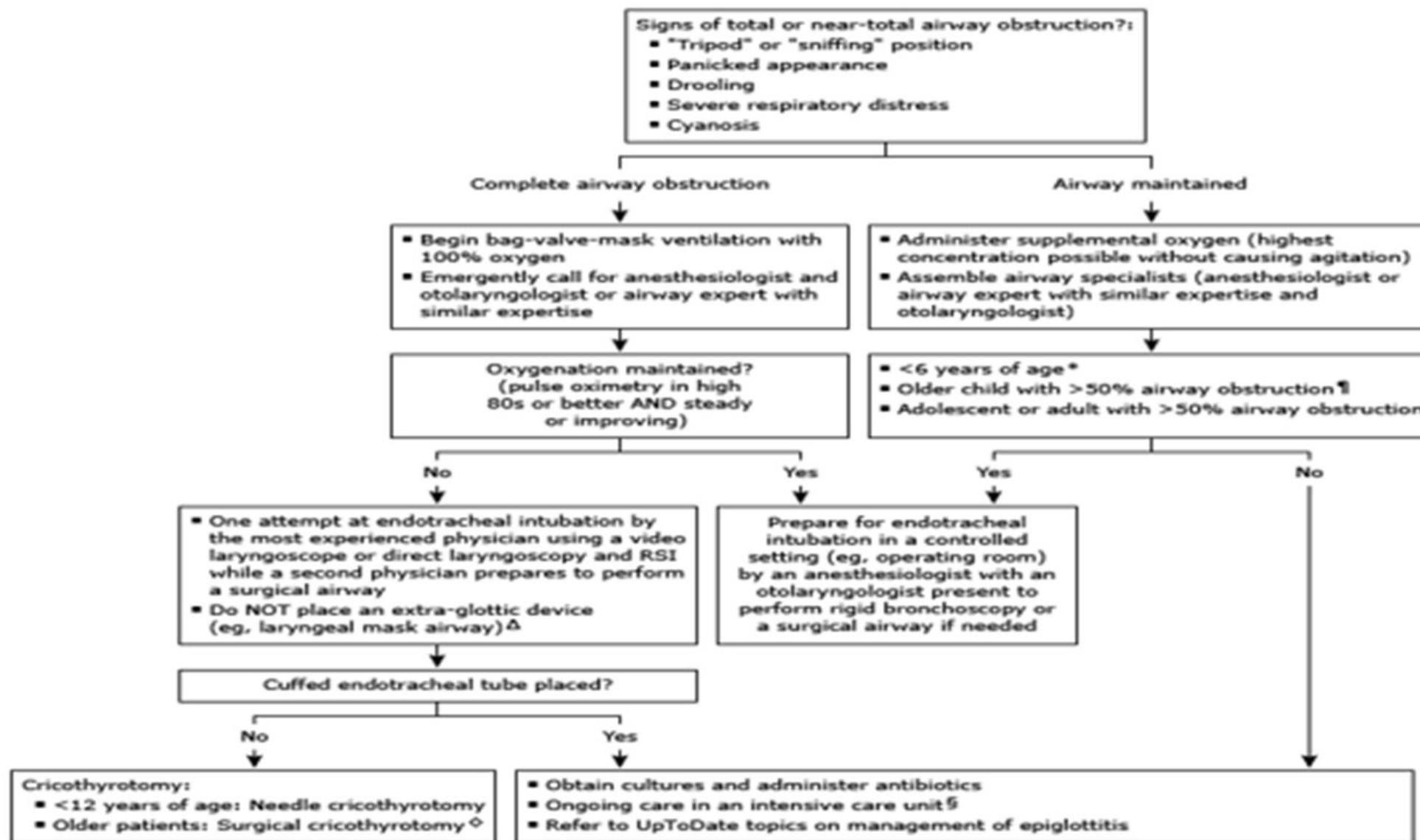
DDX

- ❖ Croup
- ❖ Bacterial tracheitis
- ❖ Peritonsillar or retropharyngeal infection
- ❖ Foreign bodies
- ❖ Angioedema (anaphylaxis or hereditary)
- ❖ Congenital anomalies and laryngeal papillomas
- ❖ Diphtheria
- ❖ Other causes of epiglottic enlargement
- ❖ Uvulitis

Management

- Immediate airway management
- Supplemental oxygen
- Antibiotics
- Supportive measure
- Minimize disturbance

Initial airway management for patients with epiglottitis



Complication

- Airway obstruction
- Epiglottic abscess
- Secondary infection
- Necrotizing epiglottitis
- Death

PREVENTION

- Immunization
- Postexposure prophylaxis

Croup (Laryngotracheobronchitis)

- Croup is a respiratory illness characterized by inspiratory stridor, barking cough, and hoarseness.
- These symptoms result from inflammation in the larynx and subglottic airway.
- Croup is one of the most common respiratory illnesses in young children.
- It occurs mostly in children ≤ 6 years old, with a peak incidence between six months to three years of age; it is uncommon in children >6 years old

Etiology

VIRAL

- Parainfluenza viruses(T1,2,3)
- Rhinovirus, respiratory syncytial virus (RSV), and adenoviruses
- SARS-CoV-2
- Influenza
- Other human coronaviruses
- Measles

BACTERIAL

- Mycoplasma pneumoniae,
- Corynebacterium diphtheriae.
- Secondary bacterial infection

Pathogenesis

- The viruses that cause croup typically infect the nasal and pharyngeal mucosal epithelia initially and then spread locally along the respiratory epithelium to the larynx and trachea.
- Anatomic hallmark of croup is narrowing of the subglottic airway, the portion of the larynx immediately below the vocal folds.
- The cricoid cartilage of the subglottis is a complete cartilaginous ring, unlike the tracheal rings, which are horseshoe shaped.

Clinical presentation

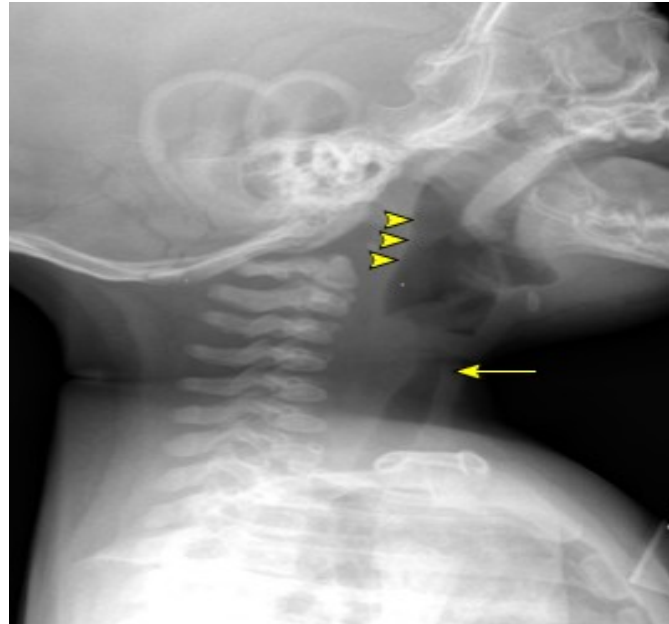
- Symptoms usually begin with nasal discharge and congestion and progress over 12 to 48 hours to include fever, hoarseness, barking cough, and stridor.
- As airway narrowing progresses, stridor develops and there may be mild tachypnea with a prolonged inspiratory phase.
- When airway obstruction becomes severe, suprasternal, subcostal, and intercostal retractions may be seen , Breath sounds can be diminished.

Cont...

- Underlying host factors that predispose to clinically significant narrowing of the upper airway include:
 - Congenital and acquired airway abnormalities,
 - laryngomalacia, laryngeal clefts, subglottic stenosis, vallecular cysts, or subglottic hemangiomas
 - Hyperactive airways,
 - Previous intubation

Diagnosis

- Croup is diagnosed clinically, based upon the characteristic barking cough and stridor
- Supportive diagnosis via radiography and laboratory



Severity assessment

- Mild croup (Westley croup score ≤ 2)
- Moderate croup (Westley croup score 3 to 7)
- Severe croup (Westley croup score ≥ 8)
- Impending respiratory failure (Westley croup score ≥ 12)

Points	0	1	2	3	4	5
Inspiratory Stridor	Not present	When agitated/active	At rest			
Intercostal Recession	No recession	Mild recessions	Moderate	Severe		
Air entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With agitation/activity	At rest
Conscious level	Normal					Altered

Differential diagnosis

- ❖ Acute epiglottitis
- ❖ Bacterial tracheitis
- ❖ Peritonsillar, parapharyngeal, or retropharyngeal abscesses
- ❖ Foreign body
- ❖ Allergic reaction or acute angioneurotic edema
- ❖ Upper airway injury
- ❖ Anomalies of the airway

Management

- ☐ Depend on severity of croup
- ☐ Mild croup → out patient with oral glucocorticoid
- ☐ Moderate to severe croup
- ☐ Managed at EOPD
 - ☐ Airway management
 - ☐ Supplemental oxygen
 - ☐ Iv/im glucocorticoid
 - ☐ Nebulized epinephrin
 - ☐ Hydration
 - ☐ Supportive care

OUTCOME

- Croup is usually a self-limited illness that resolves within 24 to 72 hours, though the cough may persist for up to one week
- Approximately 5 to 10 percent of children with croup presenting to the emergency department require hospitalization.
- Mortality is exceedingly rare (<0.1 percent of those who require hospitalization)

Reference

Nelson text book of pediatrics 22nd edition

Rudolph's pediatrics 23rd edition

Red book 2024

AAP 2018

UpToDate 2024

THANK YOU!!!